SpaCal-Depth: Guaranteed Per-Pixel Depth Intervals for Frozen Monocular Models and Sparse Depth Measurements

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Abstract

Monocular metric depth foundation models predict a distance in metres at every pixel of a single image, but they do not say reliably how wrong each prediction may be. This project attaches per-pixel depth intervals with finite-sample coverage guarantees to such models without retraining them, using conformal prediction. The work has two parts. The first is a broad study over four frozen models (Depth Anything V2, UniDepthV2, Depth Pro, Metric3D v2) and five datasets (NYU Depth v2, KITTI, DIODE, ETH3D, ARKitScenes). It shows that the image or the scene, not the pixel, is the exchangeable unit. Most per-image miscoverage comes from one global scale error per image, and a few sparse metric depth measurements (*anchors*) remove most of it, giving intervals 1.1–3.7$\times$ narrower. However, when the anchors follow a realistic sensor layout (scan lines, one cluster, only near surfaces) and are reused for calibration inside the image, the per-image guarantee fails silently: under clustered anchors, coverage falls to 0.31–0.47 at a 0.9 target. The second part, the main contribution, separates the two roles of the measurements. Structured anchors build the prediction, and a few uniformly random *probes*, never seen by the predictor, set one conformal threshold per image (A-dual). This gives a per-image coverage guarantee for any anchor layout and any probe-independent centre, with a known Beta law for its spread across images, and it needs no calibration images. On NYU and KITTI it reaches its theoretical coverage in every layout. On ARKitScenes, with iPad LiDAR checked against laser scans, it runs unchanged on a depth-completion network. That network is the better centre under uniform anchors, while the frozen model with our anchor-built centre is better under clustered anchors. The probes certify what they measure: laser probes certify the depth, while iPad probes certify only the iPad reading.

1 Introduction

Single-image depth estimation has moved from relative depth, known only up to an unknown scale and shift [15, 41], to *metric* foundation models trained on large mixtures of data. Depth Anything V2 [56], UniDepthV2 [37], Metric3D v2 [25] and Depth Pro [10] output metres zero-shot. An augmented-reality device placing content on a table, or an indoor robot planning around furniture, can run such a model, but it also needs to know how far to trust each pixel. Native uncertainty signals are not reliable. In our measurements, flip disagreement and ensemble spread cover only 3–43% of pixels when they claim 90% (Section 5.2), and the native confidences of feed-forward 3D reconstruction models are overconfident [13].

Conformal prediction [1, 29, 52] offers a principled alternative. Given a held-out calibration set that is exchangeable with the test data, it turns any predictor into intervals that cover the truth with a user-chosen probability, $1 - \alpha$, in finite samples and without distributional assumptions. This project

Course project report. [TODO: course name, code and semester]

asks what it takes to obtain such guarantees for dense metric depth from a frozen model, and what kind of guarantee is worth having.

Two parts. The project proceeded in two phases. The first (experiments E0–E16) is a broad study of conformal wrappers for frozen depth models: four models, five datasets, nine difficulty scores and ten calibrators, followed by a study of sparse test-time depth measurements (*anchors*) under realistic sensor layouts and sensor noise. Its findings motivate the second phase (E17–E20), the main contribution: a calibration layer that gives a per-image guarantee even when the measurements are placed by a structured sensor.

From the broad study to the main method. Three findings of the broad study shape the method (Section 5.2). (i) Pixels are not exchangeable; the image or the scene is the unit. (ii) A guarantee that averages over images hides individual images with large errors, mostly one global scale error per image, which about 20 sparse metric anchors remove. (iii) Calibrating inside each image on its own anchors gives a per-image guarantee, but only if the anchors are a uniform random sample of pixels. When anchors placed by a real sensor (scan lines, one cluster, only near surfaces) both correct the prediction and calibrate it, nearby anchors agree, the calibration sees small errors, and the intervals become *narrower* while missing most of the image: coverage falls to 0.31–0.47 under clustered anchors at a 0.9 target. Only dense ground truth reveals this failure.

Main idea. We give the sensor budget two roles (Fig. 1). *Prediction anchors* follow the sensor layout, whatever it is, and build a local centre and scale around the frozen prediction. *Calibration probes*, a few measurements placed uniformly at random and never shown to the predictor, set one conformal threshold per image. Because centre and scale are fixed before the probes are revealed, a rank argument gives a per-image coverage guarantee, in expectation over the probe draw, for any anchor layout and any centre that ignores the probes, including a depth-completion network. No calibration images are needed. The spread of per-image coverage follows an exact beta-binomial law, so the number of probes is a design parameter that trades sensor budget for per-image reliability.

Contributions.

- **Broad study (foundation):** a post-hoc conformal wrapper for frozen metric depth models with average, scene-level, image-event, patch and group-conditional guarantees. It shows that the image or scene is the exchangeable unit and that one scale error per image dominates. It also evaluates four anchor methods and a distance-aware one (A-dist) under structured layouts, layout mismatch, noisy and real LiDAR anchors, and a downstream safety query (Section 5.2).
- **Failure:** reusing structured anchors for calibration voids the per-image guarantee, silently and with narrower intervals (Section 5.3).
- **Dual-role calibration layer (main contribution):** A-dual gives a per-image guarantee for any layout and any probe-independent centre, with a known per-image coverage law; A-dual-both adds a guarantee by distance to the nearest anchor (Section 4.3).
- **Which centre, and what is certified:** on ARKitScenes [7], with iPad LiDAR checked against laser scans, a confirmatory comparison of a depth-completion network [57] with the frozen model as centre, and the finding that probes certify the quantity they measure (Sections 5.4 and 5.5).

The statistical ingredients are known: fitting and calibration sets inside one instance [23], within-field exchangeability [33], and the conditional coverage law of split conformal prediction [51]. Our contribution is their composition for sparse-guided dense depth, the analysis of the sensor budget, and an evaluation with an independent sensor.

Status of the numbers. The project ends with a confirmatory replication whose analysis plan was fixed before the runs (Section 5.1). All of it is complete: the replication on NYU/KITTI (E17c), the depth-completion comparison (E19c), the replication of the sensor study (E18c) and the sensor-error envelope (E20). Numbers printed in orange are exploratory and were not replicated (motivation numbers, the probe-count ablation and OMNI-DC from iPad anchors). Numbers from the broad study (E0–E16) are final for that study and printed in black.

2 Literature Review

Monocular metric depth foundation models. Learned single-image depth started with multi-scale regression networks [15]. For a long time it was relative: MiDaS [41] predicts depth up to an unknown scale and shift, recovered per image by least squares against a few known depths. Large training mixtures and camera-aware designs have since made it metric and zero-shot. Depth Anything V2 [56] provides metric fine-tunes for indoor and outdoor scenes; UniDepth and UniDepthV2 [36, 37] predict camera and depth jointly; Metric3D v2 [25] predicts in a canonical camera space and rescales by the focal length; Depth Pro [10] estimates the field of view itself. We use four of these as frozen black boxes. Their remaining error has a large per-image scale component (Section 5.2), the same component that MiDaS-style alignment removes.

Depth completion and sparse-guided depth. Depth completion predicts dense depth from RGB and sparse depth. Approaches range from early learned models [32, 48] and spatial propagation networks [12, 35] to recent zero-shot methods. OMNI-DC [57] is trained on synthetic data with diverse sparse patterns; Prompt Depth Anything [31] conditions a foundation model on a low-resolution LiDAR map; Marigold-DC [50] guides a diffusion depth prior with sparse points; Prior Depth Anything [53] fuses metric priors of any sparsity with a relative-depth model. VI-Depth [55] and MRAC [42] align a frozen monocular prediction to sparse metric depth before a learned refinement. All of these output point estimates. Uncertainty-aware completion [17, 40] learns a per-pixel uncertainty but gives no coverage guarantee. Our layer does not compete with completion methods: it wraps whichever centre is best, and we run it on OMNI-DC as well as on frozen models.

Uncertainty for depth. Bayesian, ensemble, self-supervised and gradient-based uncertainties [24, 27, 28, 39] rank pixels by difficulty, without a guarantee that a nominal 90% interval covers 90% of pixels. We use several of them as *scores* inside conformal calibration (Section 4.1). Native confidences of feed-forward 3D reconstruction models are overconfident, and after a per-model repair about two thirds of held-out scenes still miss nominal coverage by more than five points [13].

Conformal prediction and risk control. Split conformal prediction [1, 29, 52] computes a quantile of nonconformity scores on a calibration set. If calibration and test points are exchangeable, the resulting interval covers a new point with probability at least $1 - \alpha$. Conformal risk control (CRC) [3] extends this from miscoverage of one point to the expectation of any bounded monotone loss, such as the fraction of an image's pixels left uncovered. Risk-controlling prediction sets [8] make the same statement with high probability over the calibration draw. When data are grouped (pixels in images, images in scenes), the group is the natural exchangeable unit, and hierarchical conformal methods [14] treat this case. Our scene-level calibrators are of this kind.

Conformal prediction for dense outputs. Image-to-image regression [2], K-RCPS [46] and conformal segmentation [34] calibrate across images, so their guarantees average over images. Kandinsky conformal prediction [4, 11] pools scores of similar regions across calibration images, a guarantee between image-wise and pixel-wise. Online risk control [18] bounds the long-run miscoverage over a stream; its depth experiments fit each frame's scale and shift to a few hundred true depth pixels, a precedent for anchor rescaling. Distribution-free conditional coverage is impossible in general [5, 51]. Mondrian calibration [52], localized conformal prediction [22] and covariate-class guarantees [21] recover it for chosen groups, as our depth-binned and distance-binned calibrators do. Without exchangeability, coverage degrades with the distance from it [6] or must be restored by reweighting [47]; structured anchors are such a departure, and we restore exchangeability by design instead.

Conformal prediction inside one instance. A per-image guarantee needs calibration data from the test image itself. Conformalized matrix completion [23] splits the observed entries of one matrix into a fitting set and a calibration set; this is the statistical template of our dual roles. Within one spatial field, randomly sampled locations are exchangeable with unsampled ones [33]. Local spatial conformal prediction (LSCP) [26] builds intervals from nearby observations, and design-based conformal prediction [54] ties validity to the sampling design. These methods assume a known random sampling design; we face a sensor that imposes a structured one, and we spend a small random budget to restore a known design. With noisy calibration labels, conformal sets cover the

noisy label, which is conservative for the clean label only under dispersive noise [16]. Sensor-valued probes are noisy labels with a systematic bias (Section 5.5).

Decisions and sensing. For safety-critical use, the guarantee should concern the quantity a decision depends on. Simultaneous coverage of every pixel is excessive for this purpose [30], a point our safety-query experiment measures on depth (Section 5.2). Sample placement has been optimized for completion accuracy, end-to-end [9] or from uncertainty [45], and steerable sensors make random sampling possible [38]. We study given layouts and propose no placement strategy.

3 Problem Statement

Prediction and residuals. An image x has a finite set $\mathcal{P}$ of pixels with valid ground-truth depth $y(p) > 0$. A frozen monocular metric model f predicts $\hat{y}(p) = f(x)(p)$; nothing is retrained. We work with the log-depth residual

$$d(p) = \log y(p) - \log \hat{y}(p), \tag{1}$$

so an interval that is symmetric in d is multiplicative in depth: $C(p) = [\hat{y}(p)e^{m(p)-w(p)},\ \hat{y}(p)e^{m(p)+w(p)}]$ for a centre shift m and a half-width w.

A primer on conformal prediction. Let $R = \mathcal{R}(x, y)$ be a nonconformity score, *e.g.* $|d|/s$ for a scale map s, computed on N calibration units and on one test unit. If the $N + 1$ scores are exchangeable (their joint law is invariant under permutations; i.i.d. is a special case), the test score is equally likely to take any rank among them. Hence the $\lceil (N + 1)(1 - \alpha) \rceil$-th smallest calibration score $\hat{q}$ satisfies $\mathbb{P}(R_{\text{test}} \leq \hat{q}) \geq 1 - \alpha$. The interval $\{y : \mathcal{R}(x, y) \leq \hat{q}\}$ therefore covers the truth with probability at least $1 - \alpha$, for any model and any data distribution. This is *split conformal prediction*; its only assumption is exchangeability. For dense outputs, conformal risk control [3] chooses $\hat{q}$ so that the expected value of a bounded loss, such as the fraction of a new image's pixels left uncovered, is at most α.

Why pixels are not the unit. If a model misjudges the size of a room by 20%, every pixel in that image is wrong together. Pixels of one image are therefore far from independent, and treating them as exchangeable calibration points gives no valid statement. The exchangeable unit is the *image*, or the *scene* (room, drive, visit) when several images share one. A guarantee built this way holds *on average over images*. Individual images can still be covered anywhere from 40% to 100%, because each image has its own scale error.

Sparse metric measurements. A sensor returns k measurements per image. At a measured pixel it reports a depth $\tilde{y}(p)$ and a residual $\tilde{d}(p) = \log \tilde{y}(p) - \log \hat{y}(p)$. For a perfect sensor $\tilde{d} = d$; for a real one $\tilde{d}(p) = d(p) + e(p)$ with a sensor error e. The positions follow a *layout*, a random rule that may depend on x and $\hat{y}$ but never on y. We use five: *uniform* (a uniform sample of $\mathcal{P}$), *rows* (pixels on four random image rows, a few-beam scanner), *cluster* (the k pixels nearest one random centre, a single observed patch), *edges* (uniform among edge pixels of $\hat{y}$, like feature points) and *near* (uniform among the 30% of pixels with the smallest $\hat{y}$, a range-limited sensor). In the main method, $k - n$ measurements follow the layout and form the *structured anchor* set $\mathcal{S}$. The remaining n form the *probe* set $\mathcal{Q}$, drawn uniformly without replacement from $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$. Intervals are evaluated on $\mathcal{E} = \mathcal{U} \setminus \mathcal{Q}$. For a pixel p, $\delta(p)$ is its distance to the nearest measurement, divided by the image diagonal. Random probes need a sensor that can be read at random pixels: a dense or semi-dense sensor subsampled in software (phone LiDAR, time-of-flight or structured-light cameras), or a steerable scanner [9, 38].

Coverage targets. A method outputs $C(p) = [\ell(p), u(p)]$ at every evaluation pixel. The coverage of image i is

$$\text{Cov}_i = \frac{1}{|\mathcal{E}_i|} \sum_{p \in \mathcal{E}_i} \mathbf{1}\{y(p) \in C(p)\}. \tag{2}$$

At level $1 - \alpha$ we distinguish the following targets:

- *Marginal (image or scene):* $\mathbb{E}[\text{Cov}_i] \geq 1 - \alpha$ over images (or scenes), measurements and calibration data.

- *Image event:* with probability at least $1-\alpha$, a new image has at least a fraction $1-\beta$ of its pixels covered.
- *Per-image:* $\mathbb{E}[\mathrm{Cov}_i \mid x_i, \mathcal{S}_i] \geq 1-\alpha$ for *every* image, with only the probes random. No image is then always under-covered.
- *Per distance bin:* for bins of δ (edges $0.02, 0.05, 0.1, 0.2$), the expected coverage of the pixels of a new scene in each bin is at least $1-\alpha$, so far pixels are not traded for near ones.

Every guarantee concerns the values used to calibrate: sensor readings certify the readings, not the depth. We always *evaluate* against independent ground truth y.

Efficiency. Among valid methods we prefer narrower intervals. We report the median relative width $(u-\ell)/\hat{y}$ and the interval score $(u-\ell) + \frac{2}{\alpha}[(\ell-y)_+ + (y-u)_+]$, which also penalizes misses by their size. The *relative interval score* (IS) divides the interval score by y, averages over the pixels of an image, and takes the median over images. It keeps the miss penalty while staying robust to a few extreme images.

Research questions. (Q1) Which unit makes conformal guarantees for dense depth valid, and what do they cost? (Q2) How much do sparse metric anchors help, and do their guarantees survive realistic layouts, layout mismatch and sensor noise? (Q3) Can a per-image guarantee be kept for *any* layout and any centre, and what does the sensor budget buy? (Q4) Which centre should be calibrated, and what exactly is certified when the measurements come from a real, imperfect sensor?

4 Proposed Method

The method grew in three stages, each answering a failure of the previous one. Section 4.1 wraps a frozen model in cross-image conformal calibration (broad study). Section 4.2 adds sparse test-time anchors and a distance-aware model of their information (broad study). Section 4.3 presents the main method, which separates the anchors' two roles. Table 6 in Appendix B summarizes which method gives which guarantee.

4.1 A conformal wrapper for frozen depth models

All intervals have the form $\hat{y}\, e^{\pm\lambda s(x)}$ (shifted by m where anchors are available), with a *scale map* s that encodes how difficult each pixel is, and a threshold λ set by calibration.

Scale maps and calibrators. We compare nine scale maps: constant error in metres, error proportional to $\hat{y}$, constant relative error (the *plain log score*, $s \equiv 1$), flip disagreement, ensemble spread, gradient-based uncertainty [24], the model's own confidence map, a *learned* score (gradient-boosted trees fitted on held-out *head-train* images), and a *factorised* score $a(\text{image}) \times b(\text{pixel})$ that separates "how wrong is this image" from "where in it". With images as units, the calibrators (Table 7 in Appendix D) are: K1, conformal risk control on the fraction of an image's pixels left uncovered (marginal coverage over images); RCPS [8], its high-probability version; V4, split conformal on each image's $(1-\beta)$-quantile of scores (the image event); K2, every pixel of a fixed patch covered at once; K3, Mondrian per predicted-depth bin or for edge versus smooth pixels; and K0, a naive baseline that treats pixels as exchangeable. Scene-level versions (K1-scene, V4-scene, K2-scene) weight scenes equally and are valid when scenes, not images, are exchangeable; they need at least $1/\alpha - 1$ calibration scenes.

4.2 Sparse test-time anchors

Each test image comes with k anchors, and calibration images receive anchors by the same rule. In the broad study all k measurements play both roles.

- A-rescale shifts the prediction by the median anchor residual, $m = \mathrm{med}_{p\in\mathcal{A}}\, \tilde{d}(p)$, then calibrates across scenes (K1-scene). It gives a marginal guarantee.
- A-hybrid additionally divides each pixel's score by the image's anchor-estimated spread (leave-one-out median absolute deviation plus a floor τ from head-train). It gives a marginal guarantee.

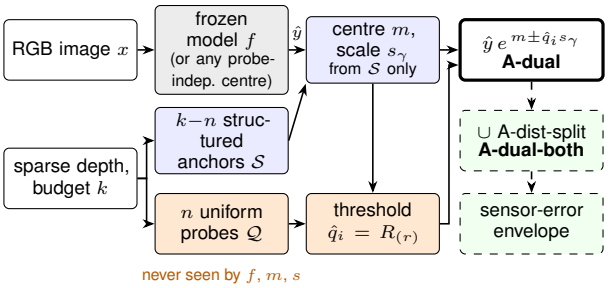


Figure 1: **Dual-role calibration.** Structured anchors $\mathcal{S}$ build a centre m and scale s_γ around a frozen prediction $\hat{y}$ (or any probe-independent centre); the n uniform probes $\mathcal{Q}$ only score it and set one threshold per image (Theorem 1). Dashed: optional stages (Theorem 2; Supp. D).

- A-local calibrates inside the image only: half of the anchors fix the shift, the other half set the width by split conformal [33]. If the anchors are a uniform sample of pixels, it gives a per-image guarantee with exact value $r/(k/2+1)$, $r = \lceil (k/2+1)(1-\alpha) \rceil$, and no calibration set is needed. It is infinite for $k < 18$ at $\alpha = 0.1$.
- A-both is the pointwise hull of A-hybrid and A-local; since both statements are monotone in the interval, it inherits both guarantees [19].

Distance-aware anchors (A-dist). Under structured layouts, coverage depends strongly on the distance to the nearest anchor (Section 5.2). A-dist models this. The local shift $m(p)$ is one of four estimates from the anchor residuals: the image median, the median of the 16 nearest anchors, or a Gaussian or bilateral (also weighting similarity in $\log \hat{y}$) kernel mean. The local spread $\sigma(p)$ is the median absolute deviation of the 16 nearest anchor residuals, and the scale is

$$s(p) = \big(\sigma(p) + \tau\big)\, g\big(\delta(p)\big), \tag{3}$$

with a floor τ and a non-decreasing g fitted by isotonic regression of normalized error against distance. The variant and bandwidths are chosen per layout on head-train images ("auto"), never on calibration or test data. One threshold per bin of δ is then set by CRC over calibration scenes, with the miscovered fraction of bin pixels as loss. This is a Mondrian construction: the expected bin miscoverage of a new scene is at most α in every bin, and sparse bins are merged. In the main method this calibrated version is called A-dist-split, and the same anchor model supplies the centre of A-dual.

Layout mismatch. Cross-image calibration is valid only if calibration and test anchors follow the same layout. If the deployed layout is one of a known set, calibrating under each and keeping the largest threshold ("@max") is valid for each of them: for every layout, the @max threshold is at least that layout's own valid threshold. A random mixture of layouts at calibration time is valid only on average over that mixture.

4.3 Main method: dual-role anchors (A-dual)

Architecture (Fig. 1). (1) A frozen monocular model produces $\hat{y}$. (2) The sensor budget is split into $k - n$ anchors $\mathcal{S}$ that follow the deployed layout and n uniformly random probes $\mathcal{Q}$. (3) The anchors alone build a local centre $m(p)$ and scale $s(p)$ around $\hat{y}$ with the A-dist model of Eq. (3), fitted on $\mathcal{S}$ only. (4) The probes, unseen by stages 1–3, set one conformal threshold per image. Stage 1 can be any predictor that does not see the probes, *e.g.* a depth-completion network fed RGB and $\mathcal{S}$; stage 3 is then refit around its output. A-dual is therefore a calibration layer, not a depth estimator.

Scale shrinkage. When the anchors set the shift wrong everywhere, as a single cluster can, the residual is nearly flat. The threshold is then set by the probes with the smallest s, and widths elsewhere grow by up to $\max s / \min s$. We therefore shrink the scale toward its per-image median $M_i = \mathrm{med}_{p \in \mathcal{P}_i}\, s(p)$:

$$s_\gamma(p) = M_i \big(s(p)/M_i\big)^\gamma, \qquad \gamma = 0.5. \tag{4}$$

Table 1: **Probe budget at $\alpha = 0.1$, one probe draw**, from the Beta law of Proposition 1 alone (no data).

n	r	$r/(n+1)$	10th percentile of Cov_i	$\mathbb{P}(\mathrm{Cov}_i < 0.85)$, %
10	10	0.909	0.795	19.7
20	19	0.905	0.819	17.6
40	37	0.902	0.841	13.0

Per-image threshold. For probes $\mathcal{Q} = \{p_1, \ldots, p_n\}$ with reported residuals $\tilde{d}(p_j)$,

$$R_j = \frac{|\tilde{d}(p_j) - m(p_j)|}{s_\gamma(p_j)}, \qquad \hat{q}_i = R_{(r)}, \qquad r = \lceil (n+1)(1-\alpha) \rceil, \tag{5}$$

where $R_{(r)}$ is the r-th smallest score ($\hat{q}_i = \infty$ if $r > n$). The interval at every evaluation pixel is

$$C(p) = \left[\hat{y}(p)\, e^{\, m(p) - \hat{q}_i s_\gamma(p)},\ \hat{y}(p)\, e^{\, m(p) + \hat{q}_i s_\gamma(p)} \right]. \tag{6}$$

Theorem 1 (Dual-role validity)**.** *Fix an image with valid pixels $\mathcal{P}$ and a structured anchor set $\mathcal{S} \subset \mathcal{P}$ chosen by any rule, which may depend on x, $\hat{y}$ and the anchor values. Let m, $s > 0$ and γ be determined by $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$. Let v be the log residual of the quantity the probes report ($v = d$ for exact depth, $v = \tilde{d}$ for a sensor). Draw $\mathcal{Q}$ uniformly without replacement from $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$, compute $\hat{q}$ by Eq. (5) with v, and let J be uniform on $\mathcal{U} \setminus \mathcal{Q}$. Then*

$$\mathbb{P}\big(\hat{y}(J) e^{v(J)} \in C(J) \,\big|\, x, \mathcal{S}, \tilde{d}|_{\mathcal{S}}\big) \ \geq\ \frac{r}{n+1} \ \geq\ 1 - \alpha,$$

with equality on the left when the scores have no ties and $r \leq n$. Equivalently, $\mathbb{E}[\mathrm{Cov}_i \mid x, \mathcal{S}] \geq r/(n+1)$, the expectation being over the probe draw.

Proof idea. Given the anchors, every pixel of $\mathcal{U}$ carries a fixed score. The probes together with J form a uniformly random $(n+1)$-subset of $\mathcal{U}$, so J's score is among the r smallest of the $n+1$ scores, and hence covered, with probability at least $r/(n+1)$. The full proof is in Appendix A.1. □

The theorem needs no calibration images and uses all n probes for the threshold; $n \geq 9$ suffices at $\alpha = 0.1$. By contrast, A-local-split, which applies A-local to the n probes, spends half of them on a shift. *Scope:* the guarantee holds in expectation over the probe draw, not per region. Probes must be uniform over the pixels it concerns. Every choice, γ included, must be fixed without looking at probe values. And it covers v, the quantity the probes measure.

Probe budget. Theorem 1 fixes the mean of Cov_i; the number of probes fixes its spread.

Proposition 1 (Per-image coverage law)**.** *Under Theorem 1 with no ties and $r \leq n$, let $N = |\mathcal{U}|$. The number of covered evaluation pixels, $(N-n)\mathrm{Cov}_i$, follows exactly the beta-binomial law* $\mathrm{BetaBin}(N-n,\, r,\, n+1-r)$. *As $N \to \infty$, $\mathrm{Cov}_i \to \mathrm{Beta}(r,\, n+1-r)$ in distribution.*

This is the finite-population form of the Beta law of split-conformal coverage conditional on the calibration set [51]; the proof is in Appendix A.2. It turns the probe count into a design parameter (Table 1). The share of images below coverage c in one draw, $\mathbb{P}(\mathrm{Beta}(r, n+1-r) < c)$, is known before any data are seen. At $\alpha = 0.1$ it falls slowly with n (5.5% below 0.85 at $n = 100$). The nominal level is the stronger lever: $r = n = 20$ (the A-dual-rmax arm, nominal 0.952) leaves only 3.9% of images below 0.85 (measured on NYU and KITTI: 3.7–4.0%, coverage 0.952–0.953), at 31–84% more width and 8–21% more interval score than $r = 19$.

A per-distance-bin guarantee (A-dual-both). A-dual's coverage, in practice, decreases with the distance to the nearest anchor; Theorem 1 allows this because it is an average over the image. A-dist-split (Section 4.2, fitted on all k measurements) has the opposite profile: a per-bin guarantee averaged over scenes, no per-image guarantee, and a need for calibration scenes with dense depth from the same source as the measurements. A-dual-both is the pointwise hull $C^{\text{both}}(p) = \big[\min(\ell^{\text{dual}}, \ell^{\text{dist}}),\ \max(u^{\text{dual}}, u^{\text{dist}})\big]$.

Theorem 2 (Hull validity)**.** *A-dual-both has the per-image guarantee of Theorem 1 and the per-bin guarantee of A-dist-split.*

Proof idea. The hull contains each interval pointwise and both statements are monotone in the interval [19] (Appendix A.3). □

The broad study's A-dist-both, the hull of A-dist-split with A-local-split, is the baseline that A-dual-both replaces. Any hull costs width, most when its two centres disagree, as under clustered anchors.

4.4 Sensor-error envelope

With sensor-valued probes, Theorem 1 certifies the reading $\tilde{y}$, not the depth y. Let $e = \log \tilde{y} - \log y$ and suppose a reference (here a laser scan) exists on calibration scenes. We (a) subtract a global bias $\hat{\beta}$, the median of e on head-train pixels, from the probe values. We (b) give each sensor-confidence class c (ARKit: $0, 1, 2$) the band $[-a_c, b_c] = [Q_c(\lambda/2), Q_c(1-\lambda/2)]$ of the corrected head-train error, clipped to contain 0. One conformal risk control over the U calibration scenes sets $\hat{\lambda}$: the largest value on a fixed grid with $\frac{U}{U+1}\hat{L}(\lambda) + \frac{1}{U+1} \le \alpha_e$, where $\hat{L}$ is the mean fraction of a scene's pixels outside their band. With $[\ell, u]$ the A-dual interval for the corrected reading at level $1-\alpha_s$, the enveloped interval (A-dual+env) is

$$C^{\text{env}}(p) = \left[\ell(p)\, e^{-b_{c(p)}},\ u(p)\, e^{a_{c(p)}}\right]. \tag{7}$$

Proposition 2 (Envelope)**.** *If the sensor reading and its confidence are defined at every evaluation pixel, the miscoverage of y by C^{env}, averaged over test scenes and probe draws, is at most $\alpha_s + \alpha_e$.*

Proof idea. y can be missed only if the corrected reading falls outside $[\ell, u]$ (at most α_s per image, Theorem 1) or the corrected error falls outside its band (at most α_e in expectation, by risk control); a union bound concludes (Appendix A.4). □

The per-image guarantee on y is not retained, since the envelope part holds on average over scenes. We fixed $\alpha_s = \alpha_e = 0.05$ before the run. One risk control over a single class-adaptive parameter replaces per-class thresholds, which would all be infinite with 37 calibration scenes (Appendix A.4); this design was amended before any envelope output existed.

5 Experiments and Results

5.1 Datasets, models and protocol

Datasets. Table 8 in Appendix F lists the five datasets. NYU Depth v2 [44] and KITTI [20] are used throughout; on KITTI the ground truth is built from the same LiDAR that would provide anchors, so it cannot test an independent sensor. DIODE [49] and ETH3D [43] test domain shift and the case of very few scenes. ARKitScenes [7] (depth-upsampling validation split) provides an independent sensor. Its ground truth is a Faro laser scan, while the deployed sensor is the iPad Pro LiDAR depth from ARKit (fused, with confidence classes $0, 1, 2$), which reads about 1.6% nearer than the laser. Valid pixels are kept on a regular lattice (stride 4 on NYU, 2 on KITTI and DIODE, 8 on ARKitScenes) and stored per image, so per-image statistics are exact. Every split is by scene: head-train (fits scores, m, s and hyper-parameters), calibration (cross-scene thresholds; unused by A-dual) and test. The main-method experiments use 30/35/35%.

Frozen models. We use UniDepthV2-Large [37], Depth Anything V2 Large metric [56] (indoor checkpoint on NYU, DIODE indoor and ARKitScenes; outdoor checkpoint on KITTI and DIODE outdoor), Depth Pro [10] and Metric3D v2 ViT-Large [25]. Metric3D v2 is given the calibrated focal length, while the other three infer the camera, so it is not compared like for like. NYU and KITTI are not in the training mixtures of these models. UniDepthV2's training mixture includes ARKitScenes, which can affect width but not any guarantee; Depth Pro was trained on ARKitScenes and is not used there. On ARKitScenes we also use the depth-completion network OMNI-DC [57] (v1.1, trained only on synthetic data). It sees RGB and the structured anchors only, never the probes.

Metrics. Coverage is image-averaged (Eq. (2)). Per-image spread is the 10th percentile (p10) of Cov_i and the share of *short* images, whose coverage on one probe draw is below $1-\alpha-0.05 = 0.85$ (theory at $n = 20$: 17.6%). Width is the median relative width; IS is the relative interval score of Section 3. $\alpha = 0.1$ unless stated.

Table 2: **Broad study (E0–E16), headline results.** Plain log score, $\alpha = 0.1$; ranges over the NYU/KITTI $\times$ model pairs unless stated. Values in orange are re-measured by the confirmatory run (E17c).

Experiment	Result
E0 accuracy	AbsRel on NYU / KITTI: UniDepthV2 0.071 / 0.077, DA-V2 0.206 / 0.109, Depth Pro 0.092 / 0.139, Metric3D v2 0.066 / 0.048.
E2–E4 validity	Cross-image CRC (K1) covers 0.894–0.900 on every NYU/KITTI pair; raw uncertainties claiming 90% cover 3–16% (flip), 15–43% (ensemble) and 83–91% (learned variance).
E1, E8 units	Run-to-run SD of coverage follows the Beta law with $n =$ #images and is $\approx 100\times$ the pixel-count prediction (Fig. 2).
Scene level	Image-event calibration is slightly short on scene-clustered data (0.884–0.897); its scene version meets the target (0.908–0.940) for 3–12% more width.
E5 shift, ETH3D	Calibrating on one dataset and testing on another: 33–85% coverage. On ETH3D (13 scenes) a scene-dependent scale bias drops Depth Pro to 0.69; scene-level calibration needs ≥ 9 calibration scenes and returns infinite intervals with 4.
E11 anchors	One global scale error explains 65–79% of image-to-image variation in coverage. About 20 random anchors match an oracle that knows each image's scale, giving intervals 1.1–3.7$\times$ narrower. A-local matches its exact theory (0.908–0.910 vs. 0.909 at $k = 20$), and A-both cuts short images from 16% to 0.2% (NYU, DA-V2).
E13 structured	A-local on structured anchors covers 0.31–0.47 (cluster), 0.71–0.79 (near), 0.87–0.89 (rows), with *narrower* intervals. Cross-image A-rescale and A-hybrid stay valid under every layout.
E14 A-dist	Coverage is flat across distance to the nearest anchor under all five layouts, and up to 67% narrower than A-hybrid. With 20 of the k anchors spent at random, A-dist-both covers 0.965–0.983 with $\leq$0.3% short images.
E15 mismatch	Calibrating under one layout and deploying another leaves coverage as low as 0.77 (A-hybrid, edges $\rightarrow$ cluster). The largest threshold over layouts (@max) is valid for each layout. LSCP covers 0.17–0.93, depending on the layout.
E16 noise	With matched sensor noise, every cross-image guarantee holds (0.916–0.925). Calibrating on clean anchors fails under outliers (A-dist 0.66) and a 5% bias (0.36–0.86). Within-image methods cover the sensor reading, so a bias breaks them (0.38–0.78).
E12 safety	Only guarantees calibrated on the query itself (nearest depth in a region) are valid and informative. Per-pixel guarantees leave the region minimum over-estimated in 2–24% of images, and simultaneous coverage of all pixels raises alarms on 99–100% of safe images.

Protocol and decision rules. The broad study used 20 random scene splits for the cross-image calibrators and 3–5 splits $\times$ 2–6 anchor draws for the anchor experiments. A method was called *valid* if its mean coverage over splits is at least $1 - \alpha - 2$ SE. It was called *better* than another only if the 95% paired bootstrap interval of their interval-score difference over splits excludes zero. The dual-anchor experiments were first run in an exploratory phase (3 splits $\times$ 2 probe draws), which chose the shrinkage $\gamma = 0.5$ after $\gamma = 1$ showed a heavy width tail. All comparisons are then replicated in a confirmatory run whose analysis plan was frozen before the runs: E17c (20 splits $\times$ 10 probe draws on NYU and KITTI, plus an $r = n$ arm), E18c (the same on ARKitScenes), E19c (10 visit splits, each with its own anchor mask) and E20 (the envelope). The plan also fixes two thresholds in advance. The completion centre "wins" a layout only if its best valid method beats UniDepthV2 + A-dual by at least 15% in IS. The envelope enters the main method only if its truth coverage is valid and it beats A-dist-both by at least 10%. Two amendments were logged before any affected output was seen: the primary IS definition, and the single-CRC envelope of Section 4.4.

5.2 Broad study: what conformal wrapping and anchors can and cannot do

Table 2 summarizes the broad study; details and further tables are in Appendix D.

Pixels are not the unit (Q1). On synthetic data with a controllable per-image scale error, average coverage stays at 0.90 while the spread of per-image coverage grows from 0.006 to 0.195: average validity says nothing about individual images. On real data the run-to-run variability tracks the image-count prediction within a factor of 0.8–1.7 (Fig. 2). With few, very different scenes (ETH3D,

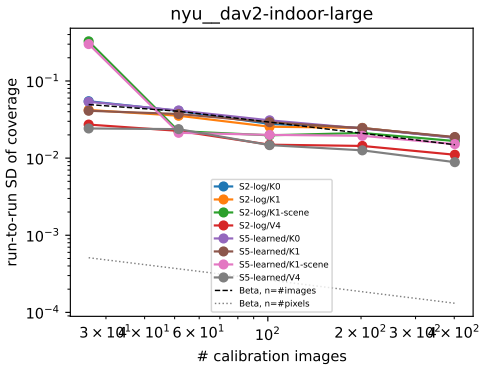


Figure 2: **The image is the unit (E8, NYU, DA-V2).** Run-to-run SD of coverage against the number of calibration images (50 splits). Dashed: Beta law with n = #images; dotted: with n = #pixels.

Reading the findings. Conformal wrapping fixes *average* coverage everywhere at modest cost, and the score matters: learned or relative scores give up to 20% lower interval score. What cross-image calibration cannot fix is the spread of coverage across individual images (16% of NYU images short with DA-V2), or a change of dataset without target calibration data. Anchors attack the dominant per-image error, the global scale. Within-image calibration then gives a per-image guarantee that needs no calibration set at all (on DIODE it is the only finite option: 0.900 coverage at $k = 200$). But it holds only for random anchors, and every failure under structured anchors is *silent*: the intervals look tighter, not looser.

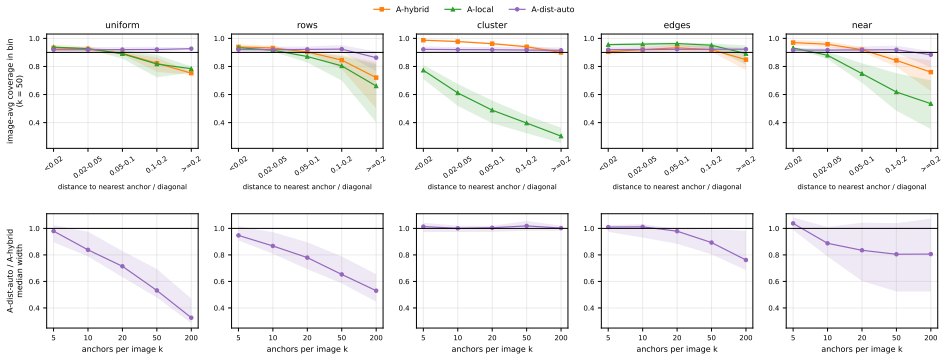


Figure 3: **Distance-aware anchors (E14).** Top: image-averaged coverage per bin of distance to the nearest anchor ($k = 50$; mean over the six NYU/KITTI pairs, band: min–max; line: 0.9). A-local collapses with distance under structured layouts; A-dist stays flat under all five. Bottom: A-dist's median width relative to A-hybrid against the number of anchors.

4 calibration scenes per split) the small slip of image-level guarantees becomes a collapse, and the valid scene-level version is uninformative.

Structured anchors break within-image calibration (Q2). Cross-image anchor methods stay valid under every layout, because calibration and test share it, at a width cost (clustered anchors make A-hybrid 26–61% wider than random anchors at $k = 20$). A-local fails, and with it A-both's per-image guarantee: under clustered or near-only anchors 5–14% of A-both's images fall below 0.80, against 0.3–2% with random anchors. Coverage also falls with distance to the nearest anchor even for random anchors, from 0.93–0.94 next to an anchor to 0.61–0.84 beyond 0.2 of the image diagonal.

The distance-aware model and its limits. A-dist removes the distance gradient (Fig. 3). At $k = 20$ under uniform anchors its farthest-bin coverage is 0.86–0.91, where A-hybrid's is 0.66–0.83. It is narrower wherever the anchors spread over the image (uniform: -29% at $k = 20$, -67% at $k = 200$;

Table 3: **NYU and KITTI, three frozen models (UniDepthV2, DA-V2, Depth Pro),** $n = 20$ **probes,** $\alpha = 0.1$ (confirmatory E17c, 20 scene splits $\times$ 10 probe draws); median over the six model–dataset pairs. local: A-local calibrated on the k structured anchors themselves; dual: A-dual; both: A-dual-both. Short: % of images whose coverage on one probe draw is below 0.85. Δ: change in median relative width and relative IS (primary) against A-dist-both. †Uniform anchors only. ‡Table 1.

		Coverage		Short (%)	Δ width (%)		Δ IS (%)	
Layout	k	local	dual	dual	dual	both	dual	both
uniform	50	0.923	0.905	17.4	−49	−37	−19	−23
	200	0.901	0.905	17.4	−69	−60	−37	−39
rows	50	0.897	0.905	17.5	−41	−30	−15	−19
	200	0.873	0.905	17.5	−52	−42	−21	−26
cluster	50	0.346	0.905	17.6	−16	−1	3	0
	200	0.405	0.905	17.6	−17	−2	2	−1
edges	50	0.959	0.905	17.6	−24	−12	−4	−8
	200	0.949	0.905	17.6	−34	−24	−8	−12
near	50	0.769	0.905	17.6	−32	−18	−7	−10
	200	0.727	0.904	17.7	−37	−22	−6	−10
theory		$\geq$0.9†	0.905	17.6‡				

rows: up to −47%) and within 6% of A-hybrid under clusters. Its guarantee is still an average over scenes, and it inherits two weaknesses of cross-image calibration. Under a layout mismatch it misses by up to 4 points on average (0.863, near → edges); @max calibration fixes this at a width cost. With noisy anchors, calibrating on clean data is unsafe: with 10% outliers it covers 0.66 at $k = 200$, and next to the anchors only 0.63. Spending 20 measurements at random and taking the hull with A-local-split (A-dist-both) restores a per-image guarantee under every layout. This budget split is the seed of the main method: the random measurements do the calibrating, so the structured ones need not.

5.3 Main method: structured anchors break reused calibration; probes restore it

Failure (Table 3, local). Calibrating within the image on the structured anchors themselves keeps nominal coverage only for uniform anchors. Under clustered anchors it covers 0.346 and 0.405 at $k = 50$ and 200; under near-only anchors 0.769 and 0.727; under scan lines 0.873 at $k = 200$. Edge anchors over-cover, since edges are where errors are largest. The failing intervals are narrower, not wider: with 20 clustered anchors their median relative width was 0.020–0.046, against 0.15–0.45 for uniform ones.

Separating the roles restores per-image validity. A-dual covers 0.904–0.905 in every layout, k and pair, at its theoretical rate $r/(n+1) = 0.905$ (within two standard errors in 56 of the 60 pair–layout–k cells). It is at least 10% narrower than the hull baseline A-dist-both on 6 of 6 pairs in each of the 10 layout–k cells, and 16–69% narrower overall (median over pairs; paired bootstrap intervals exclude 0). Part of that width is A-dist-both's over-coverage (0.964–0.985), so the interval score, which charges misses, is the fairer comparison. On it A-dual is better in 46 of the 60 cells, including every cell with uniform or scan-line anchors (by 7–42%); A-dist-both is better in 13, nine of them clustered, by at most 5%. A-dual-both is better than A-dist-both in 54 cells and worse in 4, all clustered and by at most 1.1%. Under clustered anchors its two halves are centred differently and it gains only 0.2–4.6% in width. A-dual's coverage varies across distance bins by 3.5–12.2 points; A-dual-both reduces this to 1.1–2.3 (Fig. 4a).

The probe budget. With 20 probes the 10th percentile of Cov_i was 0.819 (law: 0.819), an implementation check of the randomization (Fig. 4b). With 10 probes A-dist-both is infinite on every image while A-dual stays finite; with 40 the short share falls but A-dual misses the 10% margin over A-dist-both under clustered anchors, so we keep $n = 20$ (Appendix G).

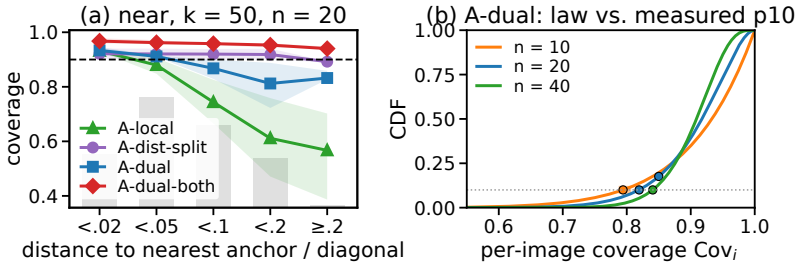


Figure 4: **What each guarantee protects (E17).** (a) Near layout, $k = 50$, $n = 20$: A-dual's coverage decreases with distance, as Theorem 1 allows; A-dual-both is flat. Lines/bands: median/range over the six NYU/KITTI pairs; bars: pixel share; dashed: 0.9. (b) CDF of the per-image coverage law (Proposition 1) for $n = 10, 20, 40$; dots at 0.1: measured p10 of Cov_i (median over pairs, layouts and k); dot at 0.85: measured share of short images, $n = 20$. Confirmatory for $n = 20$; $n = 10, 40$ exploratory.

Table 4: **ARKitScenes, which centre to calibrate (E19c, confirmatory).** $k = 200$ (180 structured anchors + 20 probes), laser-valued anchors and probes, $\alpha = 0.1$; mean over ten visit splits, one probe draw each. UniDepthV2 is frozen; OMNI-DC is completed from RGB and the 180 anchors. split-CP calibrates one quantile on dense laser depth of calibration scenes; the UniDepthV2 split-CP row uses no anchors. W/y: median width over laser depth; IS: primary relative interval score. Theory for A-dual, one draw: coverage 0.905, p10 0.819.

		Uniform layout				Cluster layout			
Centre	Calibration	Cov.	p10	W/y	IS	Cov.	p10	W/y	IS
UniDepthV2	split-CP	0.897	0.594	0.513	0.528	0.897	0.595	0.513	0.529
UniDepthV2 + $\mathcal{S}$	A-dist-split	0.923	0.860	0.027	0.082	0.924	0.837	0.071	0.135
	A-dual	0.904	0.818	0.030	0.089	0.905	0.819	0.109	0.182
	A-dual-both	0.949	0.905	0.038	0.087	0.967	0.930	0.126	0.174
OMNI-DC + $\mathcal{S}$	split-CP	0.897	0.813	0.024	0.073	0.895	0.676	0.336	0.374
	A-dist-split	0.923	0.850	0.026	0.071	0.924	0.757	0.391	0.422
	A-dual	0.904	0.819	0.023	0.074	0.905	0.820	0.226	0.311
	A-dual-both	0.948	0.902	0.032	0.074	0.971	0.917	0.403	0.421

5.4 Which centre to calibrate: a depth-completion network or the frozen model (E19c)

Uniform anchors: the completion network is the better centre. split-CP, A-dual and A-dist-split on OMNI-DC's output have a 16–20% lower IS than A-dual on UniDepthV2 (Table 4; all paired intervals exclude zero). This fires the pre-specified $\geq$15% criterion, so the layer belongs on top of the completion network. A-dual on OMNI-DC costs 2.5% in IS over split-CP on the same output and adds the per-image law: p10 0.819 and 17.8% short images, as Table 1 predicts. split-CP has no per-image guarantee and needs dense reference depth on calibration scenes. Without anchors, split-CP on the frozen model has 6–7$\times$ the IS of any anchored method.

Clustered anchors: the ranking reverses. Away from the one observed patch, the completion network extrapolates poorly. split-CP on its output is not valid (coverage 0.895, more than 2 SE below 0.9, p10 0.676). A-dual on UniDepthV2 has a 41% lower IS than A-dual on OMNI-DC, and a 51% lower IS than split-CP on OMNI-DC. On OMNI-DC itself, A-dual beats split-CP by 17%. On the frozen centre, A-dist-split is more efficient still (IS 0.135) but has no per-image guarantee, while A-dual-both keeps both guarantees at 0.174. Across both centres and layouts, A-dual's per-image spread is the same (p10 0.818–0.820, 17.4–17.8% short), as Theorem 1 requires of any probe-independent centre.

Table 5: **Laser-truth coverage by value source of anchors / probes** (ARKitScenes, $k = 200$, $\alpha = 0.1$; L = laser, I = iPad LiDAR). Frozen-model rows: confirmatory E18c, mean over 20 visit splits and ten probe draws; A-dual+env: E20, same splits and draws, iPad probes only. OMNI-DC rows: E19, laser anchors confirmatory (ten splits); iPad anchors (I/L, I/I) exploratory (three splits).

		Uniform layout				Cluster layout			
Centre	Calibration	L/L	I/L	L/I	I/I	L/L	I/L	L/I	I/I
UniDepthV2 + $\mathcal{S}$	A-dist-split	0.924	0.617	0.925	0.527	0.924	0.942	0.926	0.867
	A-dual	0.905	0.905	0.973	0.566	0.905	0.905	0.926	0.848
	A-dist-both	0.979	0.968	0.978	0.920	0.974	0.978	0.970	0.958
	A-dual+env	–	–	0.996	0.993	–	–	0.994	0.994
DA-V2 + $\mathcal{S}$	A-dual	0.905	0.905	0.969	0.632	0.905	0.905	0.920	0.877
OMNI-DC + $\mathcal{S}$	split-CP	0.897	0.897	0.897	0.897	0.895	0.894	0.895	0.894
	A-dist-split	0.923	0.545	0.923	0.539	0.924	0.922	0.924	0.922
	A-dual	0.904	0.905	0.980	0.486	0.905	0.905	0.915	0.881

Point accuracy agrees. On the test pixels outside the structured anchors (image-mean AbsRel, mean over the ten splits), OMNI-DC reaches 0.0064 under uniform and 0.0738 under clustered anchors. UniDepthV2 rescaled per image by the median $\log(y/\hat{y})$ at the same 180 anchors reaches 0.0293 and 0.0394, and 0.121 without anchors. $\delta < 1.25$ is 0.997 vs. 0.988 (uniform) and 0.944 vs. 0.981 (cluster). So under uniform anchors OMNI-DC is about $4.6\times$ more accurate, while under clustered anchors a single per-image scale on the frozen model halves OMNI-DC's error. This supports the interval-score result independently. A mean-over-images IS would instead be dominated by a few OMNI-DC pixels with extreme upper bounds under clustered anchors (Appendix E).

5.5 What the probes certify: an independent sensor (E18)

Table 5 varies what the measurements report, with coverage always measured against the laser. With laser probes, A-dual's spread on ARKitScenes, over UniDepthV2 and DA-V2 (AbsRel 0.120 and 0.466), five layouts and two budgets, is as predicted (p10 0.817–0.821). This is a check on data that were not used to choose γ.

- *iPad anchors, laser probes:* A-dual covers the laser truth at 0.905 in both layouts, with the clean-value spread (p10 0.819), because the probes measure the truth and absorb the anchors' bias into the threshold. A-dist-split, calibrated on the values the measurements report, covers it at only 0.53–0.70 under uniform anchors.
- *Laser anchors, iPad probes:* A-dual over-covers (0.969–0.973 uniform), with 2.0–2.5$\times$ wider intervals, as the iPad's own errors enter the threshold.
- *iPad values for both*, the deployed configuration without a reference: A-dual certifies the iPad reading. Under uniform anchors its intervals are about as narrow as the 1.6% sensor bias, and it covers the laser truth at only 0.566–0.632 (p10 0.11–0.18). The wider cluster intervals absorb more (0.848–0.877), and the hull A-dual-both recovers the truth there (0.95–0.96) but not under uniform anchors (0.67–0.74). Earlier within-image methods behave the same (iPad anchors: A-local 0.854, LSCP 0.548).

Two methods cover the truth in every cell. split-CP does so because it calibrates on laser depth of other scenes, which a deployed system lacks. A-dist-both does so through width, at 1.8–2.3$\times$ A-dual's IS under uniform anchors with laser values. Probes certify what they measure.

The sensor-error envelope (E20). The envelope of Section 4.4 restores truth coverage with iPad probes (0.993–0.996 over both models, layouts and anchor sources), but it over-covers: its IS is 2–78% above that of A-dist-both (uniform 2–51%, clustered 60–78%). It therefore fails the gate fixed in advance (valid truth coverage and an IS at least 10% below A-dist-both) in both layouts, and A-dist-both remains the cheaper way to cover the truth with sensor-grade probes. The union bound behind Proposition 2 is loose here: A-dual at level 0.95 already covers most of the truth, and widening every interval by the class band adds the rest again.

6 Conclusions

Findings. Conformal prediction turns frozen monocular metric depth models into per-pixel interval estimators with finite-sample guarantees, provided the guarantee is stated for the right unit: the image or the scene, not the pixel. A cross-image guarantee is valid at modest cost, but it is an average that hides images with large global scale errors, and it does not survive a change of dataset. A few sparse metric anchors remove most of the scale error (intervals 1.1–3.7$\times$ narrower). Calibrating inside each image on its own anchors gives a per-image guarantee only for uniformly random anchors; under the layouts real sensors produce, it fails silently, with tighter intervals that miss most of the image. Giving the sensor budget two roles fixes this. Structured anchors build the prediction, and a few uniformly random probes, hidden from the predictor, calibrate it within each image. The resulting layer needs no calibration images, turns the probe count into a design parameter with a known per-image coverage law, and runs on any probe-independent centre. A depth-completion network is the better centre under uniform anchors, and the frozen model with an anchor-built centre under clustered anchors (ten confirmatory splits, one network, one dataset). Laser probes certified the depth even with iPad anchors; iPad probes certified only the iPad reading.

Limitations.

- *Expectation, not every draw:* with 20 probes about 17.6% of images fall below 0.85 coverage in a single draw.
- *No spatial guarantee from probes alone:* the per-distance guarantee of A-dual-both needs calibration scenes and the same anchor rule at deployment.
- *What is certified:* the probes' quantity. The envelope that converts reading coverage into depth coverage holds only on average over scenes and, in our runs, costs more than A-dist-both. ARKit depth is fused and dense; raw sparse returns and dropouts on glass or sky are untested.
- *Selection of γ* used NYU and KITTI, so only ARKitScenes checks efficiency on held-out data, and UniDepthV2 saw ARKitScenes in training.
- *Scope:* one α in the main method, two budgets, one independent sensor and uniform probes by design. With a fixed-pattern scanner the guarantee covers only pixels with a return. There is no downstream evaluation beyond the safety-query study.
- *Efficiency under clustered anchors:* there A-dual's interval score is up to 5% above the hull baseline's; its advantage is the per-image guarantee, not the score.

Future work.

- A tighter sensor-error envelope that does not pay for both error sources separately.
- Test raw, sparse sensor returns instead of fused depth, and video, where the probes could be shared across frames.
- Study active placement of the anchors (the probes must stay random).
- Evaluate the intervals inside a downstream AR or planning task, calibrating the guarantee to the query it protects, as the safety-query study suggests.

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A Proofs

A.1 Proof of Theorem 1 (dual-role validity)

Setting. Fix an image x with its finite set $\mathcal{P}$ of valid pixels, its prediction $\hat{y}$, a structured anchor set $\mathcal{S} \subset \mathcal{P}$ and the values $\tilde{d}|_{\mathcal{S}}$ reported at $\mathcal{S}$. Everything below is conditional on $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$; since the statement holds for every realization, it also holds after averaging over any randomness in the layout. By assumption, the centre m, the scale $s > 0$, the per-image median $M = \mathrm{med}_{p\in\mathcal{P}}\, s(p)$, and hence $s_\gamma = M(s/M)^\gamma$, are functions of $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$, with γ a fixed constant. They are therefore fixed functions on $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$. Let $v : \mathcal{U} \to \mathbb{R}$ be the log residual of the quantity the probes report. Every $p \in \mathcal{U}$ then carries a fixed score $R(p) = |v(p) - m(p)|/s_\gamma(p)$. Let $N = |\mathcal{U}| \geq n + 1$ and $r = \lceil (n+1)(1-\alpha) \rceil$.

Coverage event. Since $s_\gamma > 0$ and exp is increasing, $\hat{y}(p)e^{v(p)} \in C(p)$ if and only if $m(p) - \hat{q}s_\gamma(p) \le v(p) \le m(p) + \hat{q}s_\gamma(p)$, *i.e.* if and only if $R(p) \le \hat{q}$. For $v = d$, $\hat{y}e^d = y$, so this is coverage of the true depth.

Exchangeability. Let $(Z_1, \dots, Z_{n+1})$ be a uniformly random ordered sample without replacement from $\mathcal{U}$, and set $\mathcal{Q} = \{Z_1, \dots, Z_n\}$ and $J = Z_{n+1}$. Then $\mathcal{Q}$ is a uniform n-subset of $\mathcal{U}$ and, given $\mathcal{Q}$, J is uniform on $\mathcal{U} \setminus \mathcal{Q}$, exactly as in the theorem. The law of $(Z_1, \dots, Z_{n+1})$ is invariant under permutations and R is a fixed function, so $(R(Z_1), \dots, R(Z_{n+1}))$ is exchangeable.

Rank argument. If $r > n$, $\hat{q} = \infty$ and coverage is 1. Otherwise, break ties among the $n+1$ scores uniformly at random and let ρ be the resulting rank of $R(Z_{n+1})$, which is uniform on $\{1, \dots, n+1\}$ by exchangeability. If $\rho \le r$, at most $r-1$ of the n probe scores are ranked below $R(Z_{n+1})$, so the r-th smallest probe score satisfies $\hat{q} \ge R(Z_{n+1})$. Hence $\mathbb{P}(R(J) \le \hat{q}) \ge \mathbb{P}(\rho \le r) = r/(n+1) \ge 1-\alpha$. Without ties, $R(J) \le \hat{q}$ holds if and only if $\rho \le r$, so equality holds.

From one pixel to image coverage. Given $\mathcal{Q}$, J is uniform on $\mathcal{E} = \mathcal{U} \setminus \mathcal{Q}$, so $\mathbb{P}(R(J) \le \hat{q} \mid \mathcal{Q}) = \mathrm{Cov}_i$ (with y replaced by $\hat{y}e^v$). Taking expectations over $\mathcal{Q}$ gives $\mathbb{E}[\mathrm{Cov}_i \mid x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}}] \ge r/(n+1)$. □

Remarks. (a) No assumption is made on how $\mathcal{S}$ was chosen, on the distribution of images, or on m and s beyond their independence of the probe values. This is why any layout and any probe-independent centre (*e.g.* a completion network fed RGB and $\mathcal{S}$) are allowed. (b) If any quantity used in m, s or γ were selected after looking at the probe values of the same image, R would no longer be a fixed function and the argument would fail. (c) The probes must be uniform over the pixels the guarantee concerns; if a sensor cannot return at some pixels, $\mathcal{U}$ and the guarantee are restricted to the pixels where it returns. (d) This is the split-conformal argument for a fitting set and a calibration set inside one instance [23], and the within-field exchangeability argument [33], specialized to a frozen prior and a structured fitting set.

A.2 Proof of Proposition 1 (per-image coverage law)

Assume no ties among $\{R(p)\}_{p \in \mathcal{U}}$ and $r \le n$. Sort $\mathcal{U}$ by score and let T be the position of the r-th smallest probe, so $\hat{q}$ is the T-th smallest score in $\mathcal{U}$. Exactly $T - r$ non-probe pixels have score $\le \hat{q}$, so $(N-n)\mathrm{Cov}_i = T - r$. For $T = r + j$, the sample must contain $r-1$ of the first $r+j-1$ pixels, the pixel at position $r+j$, and $n-r$ of the last $N-r-j$ pixels:

$$\mathbb{P}(T - r = j) = \binom{r+j-1}{r-1}\binom{N-r-j}{n-r}\Big/\binom{N}{n}, \qquad j = 0, \dots, N-n.$$

This negative-hypergeometric law equals $\mathrm{BetaBin}(N-n,\, r,\, n+1-r)$, a standard identity (*e.g.* via the Pólya urn), which we also checked numerically to 10^{-14} for $N = 300$, $n = 20$. Its mean is $(N-n)\,r/(n+1)$, so $\mathbb{E}[\mathrm{Cov}_i] = r/(n+1)$, the equality case of Theorem 1. A beta-binomial variable is a binomial mixture with success probability $B \sim \mathrm{Beta}(r, n+1-r)$. Dividing by $N-n$ and letting $N \to \infty$ with n fixed gives $\mathrm{Cov}_i \to B$ in distribution, the classical conditional-coverage law of split conformal prediction [51]. □

Numbers. For $n = 20$ ($r = 19$): $\mathbb{E}[\mathrm{Cov}_i] = 19/21 = 0.905$, the 10th percentile is 0.819, and $\mathbb{P}(\mathrm{Cov}_i < 0.85) = 17.6\%$ for one draw. With a few thousand evaluation pixels per image the finite-population correction is negligible at this precision.

A.3 Proof of Theorem 2 (hull validity)

The per-bin guarantee of A-dist-split. Let scenes be the exchangeable units. The U calibration scenes and one test scene are exchangeable, every image in them receives measurements by the same anchor rule, and all hyper-parameters of m and s are fitted on separate head-train scenes. For a distance bin b and threshold λ, let $L_{u,b}(\lambda) \in [0,1]$ be the average over the images of scene u of the fraction of their bin-b evaluation pixels with $|d-m|/s > \lambda$; it is non-increasing in λ. Conformal risk control [3] with $\hat{\lambda}_b = \inf\{\lambda : \frac{U_b}{U_b+1}\hat{L}_b(\lambda) + \frac{1}{U_b+1} \le \alpha\}$ gives $\mathbb{E}[L_{\text{test},b}(\hat{\lambda}_b)] \le \alpha$ for every bin (the Mondrian argument [52]). Bins supported by fewer than $2/\alpha - 1$ calibration scenes are merged

with the nearer neighbour, by a rule that depends on pixel positions only; the guarantee then holds for the merged bin.

Hull. For every evaluation pixel, $C^{\text{both}}(p) \supseteq C^{\text{dual}}(p)$ and $C^{\text{both}}(p) \supseteq C^{\text{dist}}(p)$. So the hull's coverage indicator is at least the maximum of the two coverage indicators, pointwise. Averaging over an image and taking the expectation over the probes gives Theorem 1's bound. Averaging over bin b and taking the expectation over scenes gives the per-bin bound. That A-dist-split uses the probe values in its own fit is irrelevant to the first statement, which only uses the A-dual interval. □

A.4 Proof of Proposition 2 (envelope)

Let $e = \log \tilde{y} - \log y$, $\hat{\beta}$ the bias estimated on head-train pixels, $\tilde{y}' = \tilde{y}e^{-\hat{\beta}}$ the corrected reading and $e' = e - \hat{\beta}$, so $y = \tilde{y}'e^{-e'}$. Run A-dual at level $1 - \alpha_s$ with probes reporting $\tilde{y}'$. By Theorem 1, for every test image, the expected fraction of $\mathcal{E}_i$ with $\tilde{y}' \notin [\ell, u]$ is at most α_s. The bands $[-a_c(\lambda), b_c(\lambda)]$ are quantiles of e' on head-train pixels only, so they form a fixed family that shrinks as λ grows ($\lambda = 0$: infinite bands). Hence the loss $L_u(\lambda)$, the fraction of scene u's pixels whose e' lies outside the band of their class, lies in $[0, 1]$, is non-decreasing in λ, and $L_u(0) = 0$. With exchangeable calibration and test scenes, conformal risk control (written for a loss that grows with λ) gives $\mathbb{E}[L_{\text{test}}(\hat{\lambda})] \leq \alpha_e$. If $\tilde{y}'(p) \in [\ell(p), u(p)]$ and $e'(p) \in [-a_{c(p)}, b_{c(p)}]$, then $y(p) \in [\ell(p)e^{-b_{c(p)}}, u(p)e^{a_{c(p)}}] = C^{\text{env}}(p)$. So the fraction of a test scene's pixels that miss y is at most the sum of the two failure fractions, whose expectations are at most α_s and α_e. □

Why one risk control. Conformal risk control cannot certify a level below $1/(U + 1)$, which is 0.026 with $U = 37$ calibration scenes per split. Per-class one-sided targets proportional to the class shares (1.1, 4.3 and 94.6% of pixels for confidence 0, 1, 2) would be 0.0003, 0.001 and 0.024, and every threshold would be infinite. A single two-sided risk control over one class-adaptive parameter avoids this; the grid has 400 log-spaced values in $[10^{-6}, 0.5]$.

B Which method gives which guarantee

Table 6: **Guarantees at level $1 - \alpha$.** "Per image" is in expectation over random anchors or probes and concerns the values they report. *Per distance bin, averaged over scenes. †With a reference sensor; the depth guarantee is a scene average.

Method	Stage	Marginal unit	Per image	Per bin*
K1, V4, K2, K3 (+ scene versions)	wrapper	image (scene)	–	depth bins (K3)
A-rescale, A-hybrid	anchors	scene	–	–
A-local	anchors	–	uniform anchors only	–
A-both	anchors	scene	uniform anchors only	–
A-dist-split (A-dist)	anchors	scene	–	yes
A-local-split	dual	–	yes	–
A-dist-both	dual	scene	yes	yes
A-dual	dual	–	yes, any layout	–
A-dual-both	dual	scene	yes, any layout	yes
A-dual+env	dual	scene†	sensor part only	–
split-CP (completion centre)	–	scene	–	–

C The distance-aware centre and scale

For pixel p, with anchor residuals $\tilde{d}$ at the anchor set $\mathcal{A}$ ($\mathcal{A} = \mathcal{S}$ for A-dual, all k measurements for A-dist-split):

- **Shift** $m(p)$, one of: *median* (image median of $\tilde{d}|_{\mathcal{A}}$); *knn* (median of the $K = 16$ nearest anchors); *kernel* (Gaussian-weighted mean of the 16 nearest, bandwidth $h \in \{0.05, 0.1, 0.2\}$ of the diagonal); *bilateral* (kernel weights times a Gaussian similarity in $\log \hat{y}$, bandwidth $h_d \in \{0.1, 0.3\}$).
- **Spread** $\sigma(p)$: median absolute deviation of the 16 nearest anchor residuals around their median.

- **Scale** $s(p) = (\sigma(p) + \tau)\, g(\delta_{\mathcal{A}}(p))$, with floor τ and non-decreasing g from isotonic regression of normalized error on distance, capped at its 99.9th percentile on head-train.
- **Selection**: per anchor draw, the variant and bandwidths with the best held-out log interval score on 200 head-train images with the same layout; τ, g and the cap per layout. No calibration or test data are used. In the broad study the automatic choice was a median 0.3% wider than the best fixed variant chosen after the fact, and chose the bilateral variant 55% of the time.

Distance bins: $\delta < 0.02$, 0.02–0.05, 0.05–0.1, 0.1–0.2, ≥ 0.2 of the image diagonal. A bin is merged into its nearer neighbour if fewer than 5% of head-train images have 10 pixels in it, or fewer than $2/\alpha - 1 = 19$ calibration scenes do.

D Broad-study details

Table 7: **Validity notions and calibrators of the conformal wrapper** (Section 4.1).

Promise at level $1 - \alpha$	Calibrator
none (pixels treated as exchangeable; naive baseline)	K0
expected fraction of missed pixels in a new image $\leq \alpha$	K1 (CRC)
the same for a new scene, scenes weighted equally	K1-scene
K1's statement with probability $\geq 1 - \delta$ over the calibration draw	RCPS
each fixed patch covered entirely with probability $\geq 1 - \alpha$	K2, K2-scene
K1 separately within groups (predicted-depth bins; edge vs. smooth pixels)	K3-depth, K3-edge
with probability $\geq 1 - \alpha$ a new image has $\geq 1 - \beta$ of its pixels covered	V4, V4-scene
for every test image, expected coverage over random anchors $\geq 1 - \alpha$	A-local, A-both

Anchors versus no anchors (E11; 5 splits $\times$ 6 anchor draws). Median relative width at matched average coverage, without anchors and with A-hybrid at $k = 20$: NYU 0.686→0.186 (DA-V2, 3.7×), 0.289→0.146 (UniDepthV2), 0.375→0.209 (Depth Pro), 0.268→0.166 (Metric3D v2); KITTI 0.646→0.513 (DA-V2), 0.369→0.225 (UniDepthV2), 0.716→0.335 (Depth Pro), 0.260→0.230 (Metric3D v2, 1.13×). The share of pixels where the interval claims the surface is farther than it is falls from 8.1% to 2.9% with A-both (NYU, DA-V2). On DIODE, with 3–4 calibration scans, every cross-image method is infinite, while A-local at $k = 200$ gives 0.900 coverage with 0% short images.

Structured anchors (E13; 5 splits $\times$ 3 draws). A-local coverage at $k = 20$ / $k = 200$, range over the six NYU/KITTI pairs (theory under uniform anchors: 0.909 / 0.901): uniform 0.908–0.910 / 0.901; rows 0.880–0.887 / 0.868–0.876; cluster 0.255–0.364 / 0.371–0.468; edges 0.918–0.954 / 0.925–0.960; near 0.746–0.791 / 0.708–0.760. Its median width at $k = 20$ was 0.020–0.046 under clustered anchors against 0.15–0.45 under uniform ones. Depth Pro reproduces all three E13 findings (A-local, clustered anchors, $k = 20$: 0.30 on KITTI, 0.29 on NYU).

Layout mismatch and LSCP (E15; 3 splits $\times$ 2 draws). Matched calibration covers 0.914–0.921 for all three cross-image methods. The worst mismatched calibrations average 0.852 (A-rescale, uniform → cluster), 0.774 (A-hybrid, edges → cluster, lowest pair 0.71) and 0.863 (A-dist, near → edges). Calibrating on a random mixture of layouts averages to the target over the mixture, but A-hybrid @mix covers only 0.845–0.853 on clustered test anchors. The largest threshold over layouts (@max) covers 0.915–0.979 on average over pairs at every test layout, lowest pair at least 0.89. A-dist @max is 15–52% narrower than A-hybrid @max unless the deployed anchors are clustered. LSCP [26], run in each image, has no finite-sample guarantee here and covers 0.17–0.93 depending on the layout.

Noisy anchors (E16; 3 splits $\times$ 2 draws). Sources: Gaussian noise (2%, 5%), gross outliers (5%, 10% of anchors), a 5% scale bias, and real single-scan KITTI LiDAR. With calibration anchors from the same sensor, A-rescale, A-hybrid and A-dist cover 0.916–0.925 under every source, but A-dist loses most of its width advantage under frequent outliers. Coverage of methods calibrated on clean anchors and deployed with the noisy sensor, mean (lowest pair), $k = 20$ / $k = 200$:

Source	A-rescale @clean	A-hybrid @clean	A-dist @clean
Gaussian 2%	0.917 / 0.918	0.931 / 0.939	0.923 / 0.942
Gaussian 5%	0.916 / 0.920	0.950 / 0.963	0.914 (0.87) / 0.960
outliers 5%	0.915 / 0.916	0.923 / 0.927	0.852 (0.81) / 0.764 (0.68)
outliers 10%	0.915 / 0.916	0.928 / 0.934	0.793 (0.71) / 0.662 (0.54)
bias 5%	0.862 (0.77) / 0.861 (0.76)	0.831 (0.66) / 0.815 (0.62)	0.737 (0.48) / 0.364 (0.14)

A-local and LSCP cover the sensor's reading: zero-mean noise makes them over-cover, and a 5% bias breaks them (A-local 0.78 / 0.73, LSCP 0.58 / 0.38 at $k = 20$ / 200). Real KITTI LiDAR anchors change little (A-dist 14–19% wider), but on KITTI the ground truth itself comes from LiDAR, so this is not an independent check.

Safety queries (E12; 20 splits). The query is the nearest true depth inside a fixed forward region, and inside each of its three lanes. A bound is violated when it claims the nearest surface is farther than it is. Ranges over the six pairs (thresholds 15 m on KITTI, 1.5 m on NYU; a method's own target in bold):

Method	Region min. over-est.	Some lane over-est.	Bound / true min.	False alarms
point prediction	0.41–0.94	0.51–0.98	0.98–1.20	0.01–0.13
K1 (two-sided, per pixel)	0.04–0.18	0.07–0.38	0.78–0.92	0.17–0.59
A-both, $k = 20$	0.02–0.11	0.05–0.18	0.81–0.95	0.20–0.51
every region pixel at once	$\leq$0.001	$\leq$0.001	0.24–0.45	0.99–1.00
task-scalar (region minimum)	**0.103–0.111**	0.14–0.24	0.82–0.89	0.25–0.52
lane-event (worst lane)	0.04–0.09	**0.105–0.116**	0.78–0.87	0.32–0.62

A guarantee protects only the question it was calibrated for. The averaging unit (image or scene) is part of the guarantee as well: each version of a guarantee holds for its own averaging and can fail for the other.

Metric3D v2 and ETH3D. Metric3D v2, which receives the calibrated focal length, is the most accurate model on every dataset. Its confidence map correlates positively with the error on all five datasets and is the best score on three of them, whereas UniDepthV2's confidence has no absolute scale and its correlation with the error changes sign across datasets. It is also far more robust to the domain shifts of E5 and gains least from anchors. On ETH3D, image-level K1 with Depth Pro covers 0.690 on average, about 4.5 standard errors below the target. The cause is a scene-dependent scale bias: the median per-image $\log(y/\hat{y})$ ranges from -0.13 to $+1.35$ across scenes. The valid scene-level calibrators need at least 9 calibration scenes at $\alpha = 0.1$ and return infinite intervals with 4.

E An interval-score blow-up under clustered anchors (E19c)

For OMNI-DC under clustered anchors only, the arms built on the distance model have, in some splits, extreme upper bounds on 0.1–0.4% of pixels. These arms are A-dist-split, A-dual-both, and A-dual in splits where its scale falls back to the kernel form. OMNI-DC reproduces its own input anchors, so the anchor residuals that the distance model reads are almost zero. Its spread estimate and floor collapse to about 10^{-6}, and far-bin thresholds become huge. The primary IS (a median over images) and median widths are unaffected, but a mean over images is not. This is a property of calibrating, with anchor residuals, a centre that has consumed those anchors. A-dual's probe-based threshold is far less affected.

F Datasets

G Probe-budget ablation (E17)

With 20 probes the 10th percentile of Cov_i was 0.819 (law: 0.819), and 17.2–17.9% of images were short in one draw (law: 17.6%). This is an implementation check of the randomization (Fig. 4b), not a finding. With 10 probes, A-local-split, and hence A-dist-both, is infinite on every image (5

Table 8: **Datasets.** Units are the exchangeable scenes used for splitting.

Dataset	Units	Images	Ground truth	Used in
NYU Depth v2 (labelled)	464 rooms	1,449	Kinect, <10 m	E0–E17
KITTI (Eigen test + 800 train)	61 drives	1,452	LiDAR, <80 m	E0–E17
DIODE indoor / outdoor (val.)	10 scans each	325 / 446	laser scanner	E0–E5
ETH3D (high-res. training)	13 scenes	454	laser scan	E0–E4
ARKitScenes (upsampling val.)	107 visits	5,594	Faro laser; iPad LiDAR as sensor	E0, E16–E20

calibration probes, fewer than the 9 needed), while A-dual stays finite at median width 0.202/0.178 ($k = 50/200$). In exploratory runs with 40 probes, the share of images short on average over two draws fell to 6.5–6.8% (law: 6.6%) at similar width (0.199/0.154). However, under clustered anchors A-dual then misses the 10% margin over A-dist-both, so we keep $n = 20$.

H Experiment index

Table 9: **Experiments.** Broad study E0–E16; main method E17–E20 ("c": confirmatory replication).

ID	Question	Size
E0	accuracy of the frozen models	official test subsets
E1	do guarantees behave as theory says on synthetic data; does spread follow images or pixels?	100 repetitions per setting
E2–E4, E6, E9	best scores and calibrators; conditional coverage; raw vs. conformal; α sweep	up to 17 pairs $\times$ 9 scores $\times$ 10 calibrators $\times$ 3 α $\times$ 20 splits
E5	does a guarantee survive domain shift; does recalibration restore it?	8 settings $\times$ 10 splits
E8	coverage variability vs. calibration size: images or pixels?	NYU, KITTI $\times$ 50 splits
E10	computational cost	–
E11	coverage, spread and width vs. number of random anchors	10 pairs $\times$ 8 k $\times$ 5 splits $\times$ 6 draws
E12	does the interval protect a safety query?	6 pairs $\times$ 2 α $\times$ 20 splits
E13	do anchor guarantees survive structured layouts?	6 pairs $\times$ 5 layouts $\times$ 3 k $\times$ 5 $\times$ 3
E14	distance-aware anchors; budget split	6 pairs $\times$ 5 layouts $\times$ 5 k $\times$ 3 $\times$ 2
E15	calibration–deployment layout mismatch; LSCP	6 pairs $\times$ 5 test layouts $\times$ 7 calibrations
E16	noisy and real LiDAR anchors	6 pairs $\times$ 6 sources $\times$ 2 k $\times$ 3 $\times$ 2
E17 / E17c	A-dual and A-dual-both vs. A-dist-both on NYU, KITTI; probe budget; $r = n$ arm	6 pairs $\times$ 5 layouts $\times$ 2 k; 3 $\times$ 2 / 20 $\times$ 10
E18 / E18c	laser vs. iPad values for anchors and probes (ARKitScenes)	2 models $\times$ 2 layouts $\times$ 4 cells; 20 $\times$ 2 / 20 $\times$ 10
E19 / E19c	completion network vs. frozen model as centre	2 layouts $\times$ 2 probe sources; 3 $\times$ 1 / 10 $\times$ 1
E20	sensor-error envelope	2 models $\times$ 2 layouts $\times$ 2 cells; 20 $\times$ 10

I Reproducibility

Every number in this report is a macro generated from the experiment summaries (`paper/numbers.tex`, written by `paper/tools/gen_numbers.py`). The confirmatory comparisons come from `experiments/confirm_analysis.py`, which writes the validity and paired-bootstrap tables, and the analysis plan with its amendment log is in `docs/ANALYSIS_PLAN_2026-11-03.md`. The broad-study numbers are taken from the project report (`docs/PROJECT_REPORT.md`), which also lists the scripts per experiment. **[TODO: public code link after the double-blind review period, if permitted]**